# Constraint-Aware Self-Supervised Learning for Edge Selection: Extended Abstract

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## 1 Motivation and Framework Overview

Many combinatorial problems can be formulated as graph-structured edge-selection problems requiring solutions to satisfy explicit combinatorial constraints. Because these problems are NP-hard, exact methods become prohibitively expensive as instance sizes grow, motivating efficient alternatives. Existing self-supervised learning-based approaches have emerged but still depend heavily on computationally intensive downstream processes, such as search and repair, or narrow problem-specific designs to enforce feasibility. This leaves a critical gap: there is limited support for a reusable learning framework where the learned predictions themselves provide direct, effective structural guidance to construct high-quality, feasible solutions efficiently in a single pass.

To address this gap, this work studies self-supervised learning for edge-selection optimization, where a solution is represented by a subset of edges satisfying explicit combinatorial constraints. The central idea is to train a neural model to produce edge predictions that already encode useful objective and feasibility structure, rather than using solver-generated labels or relying on heavy post-processing. The resulting predictions are then converted into feasible discrete solutions by lightweight single-pass construction. In this way, the neural model provides structural guidance, while the decoder enforces feasibility by construction.

Given an instance graph $G = (V, E)$, the model outputs a soft edge-selection matrix $P$, where $P_{ij}$ represents the learned preference for selecting edge $(i, j)$. This matrix is a continuous relaxation of the discrete edge-selection decision, making it possible to train the model with gradient-based optimization. Since no optimal solution labels are used, the loss is derived from the optimization problem itself: $\mathcal{L}(G; P) = \mathcal{L}_{\text{obj}}(G, P) + \sum_{\gamma\in\Gamma} \lambda_\gamma \mathcal{L}_\gamma(G, P)$. Here, $\mathcal{L}_{\text{obj}}$ is a differentiable surrogate of the task objective, while $\mathcal{L}_\gamma$ measures the soft violation of constraint $\gamma$. The coefficients $\lambda_\gamma$ are updated dynamically using a dual-ascent rule, increasing the weight of constraints that remain violated during training. This encourages the soft predictions to reflect both solution quality and feasibility requirements, without relying on fixed manually tuned penalty weights.

To produce these edge predictions, the graph architecture uses a cost-attention convolution (CAC), which injects edge-cost information directly into message passing. Given a cost matrix $C$, node features $r$, and a cost-sensitivity coefficient $\beta$, CAC computes a cost-aware attention matrix $W = \text{softmax}_{\text{row}}(-\beta C)$ and aggregates node information by $m = Wr$. Thus, the cost matrix defines how information flows across the graph: lower-cost edges receive larger attention weights and therefore contribute more strongly to the learned node embeddings. These embeddings are then converted into edge scores through a bilinear interaction over the node embedding matrix $H$, followed by a cost correction: $S = HW_{\text{edge}}^\top H^\top$

and $P = \mathrm{softmax}_{\mathrm{row}}(S - \alpha C)$, where $\alpha$ controls the strength of the final cost penalty. For budget-constrained problems, $\alpha$ and $\beta$ are conditioned on the available budget, enabling the model to adapt its cost sensitivity across different resource regimes.

## 2 Problem Instantiations and Empirical Findings

We instantiate the framework on three representative edge-selection problems: the Traveling Salesman Problem (TSP), Orienteering Problem (OP), and Team Orienteering Problem (TOP). TSP seeks a minimum-cost Hamiltonian cycle, OP seeks a reward-maximizing route under a travel budget, and TOP extends OP to multiple vehicles that share rewards while satisfying cross-vehicle exclusivity. These tasks differ in their objectives and constraints, but they share the same edge-selection structure: a solution is formed by selecting edges from an instance graph subject to problem-specific feasibility requirements.

Each problem is handled by the same self-supervised pipeline. First, the discrete objective is replaced by a continuous surrogate defined on the soft edge-selection matrix $P$. Second, feasibility requirements are expressed as soft constraint penalties during training. Third, the trained edge scores are passed to a single-pass decoder that enforces the hard constraints at inference time. The task-specific part of the framework lies in choosing the objective surrogate, the constraint penalties, and the decoding rule.

For TSP, the objective surrogate encourages low travel cost, while the constraint penalties encourage each city to be visited once and discourage subtours. The decoder then greedily selects high-scoring edges while enforcing degree and connectivity requirements, producing a valid Hamiltonian cycle. For OP, the objective surrogate encourages collecting high rewards, while the penalties discourage budget violations, invalid depot usage, and revisits. The decoder constructs a single route by repeatedly selecting feasible high-probability moves, while preserving enough remaining budget to return to the depot. For TOP, the same idea is extended to multiple vehicles: each vehicle constructs a budget-feasible route, and nodes already visited by earlier vehicles are masked to enforce cross-vehicle exclusivity.

Empirical evaluations across different graph scales demonstrate that the proposed framework achieves a strong quality-efficiency trade-off without relying on heavy post-processing. Across the TSP, OP, and TOP, the instantiated models are evaluated on both in-distribution setups and multiple unseen geometric layouts to rigorously assess out-of-distribution generalization. The framework consistently improves upon standard problem-specific greedy heuristics while requiring a fraction of the computational runtime demanded by exact mixed-integer programming solvers like Gurobi. Furthermore, for TSP, the search-free, single-pass protocol enables a direct comparison against representative learned baselines, including permutation policy models and adversarial generative flow networks. When these alternative learning methods are stripped of their heavy downstream procedures, such as multi-start sampling or iterative refinement, our approach achieves substantially stronger solution quality. These results suggest that the learned edge scores provide effective structural guidance directly within the neural network before decoding, and that the proposed self-supervised pipeline generalizes robustly across diverse edge-selection problem families.

The ablation studies further show that both the architecture and the feasibility-aware training objective are important. Replacing the CAC with standard graph encoders weakens the quality-efficiency trade-off, indicating that lightweight cost-aware message passing is well suited to dense edge-selection problems. Removing constraint-related loss terms also substantially degrades solution quality, showing that constraint-aware learning is important for guiding the model toward useful edge-selection structure.